

JEE MAIN REVISION MASTER: ROTATIONAL MOTION 2026 2 QUESTIONS 3.33% WEIGHTAGE

↑99.4%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Moment of Inertia & Parallel/Perpendicular Axes Theorems	12 PYQs	Inertia
Moderate	Torque, Angular Momentum & Pure Rolling	5 PYQs	Dynamics & Rolling

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Moment of Inertia & Parallel/Perpendicular Axes Theorems	$I = \int r^2 dm$, $I = I_{\text{cm}} + Md^2$, $I_z = I_x + I_y$ ($\text{laminar planar bodies}$)
Torque, Angular Momentum & Pure Rolling	$\vec{\tau} = I\vec{\alpha} = \frac{d\vec{L}}{dt}$, $K_{\text{roll}} = \frac{1}{2}Mv_{\text{cm}}^2 \left(1 + \frac{k^2}{R^2}\right)$, $a = \frac{g \sin \theta}{1 + k^2/R^2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Moment of Inertia & Parallel/Perpendicular Axes Theorems (High Yield — 12 PYQs)

Core Principle: Calculation of rotational inertia, radius of gyration $k = \sqrt{I/M}$, and axes shift theorems.

■ **Caution / JEE Pitfall:** Perpendicular axis theorem $I_z = I_x + I_y$ is strictly valid ONLY for 2D laminar (flat) objects lying in the xy-plane.

■ **Speed Trick:** Solid sphere $I_{\text{cm}} = (2/5)MR^2$; hollow spherical shell $I_{\text{cm}} = (2/3)MR^2$; circular disc $I_{\text{cm}} = (1/2)MR^2$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1 = 4\text{kgm}^2, r_1 = 20\text{cm}$, $I_2 = 20\text{kgm}^2$ and $r_2 = 30\text{cm}$..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "A body of mass 1 kg is suspended by a weightless string which passes over a frictionless pulley of mass 2 kg as shown in the figure. The mas..." [Ans: N/A]

3.2 Torque, Angular Momentum & Pure Rolling (Moderate Yield — 5 PYQs)

Core Principle: Rotational analogue of Newton's second law, conservation of angular momentum, and rolling down an incline.

■ **Caution / JEE Pitfall:** On an incline, the body with the smallest k^2/R^2 reaches the bottom first (Solid Sphere > Solid Cylinder > Hollow Sphere > Ring).

■ **Speed Trick:** In pure rolling on a flat surface, the instantaneous axis of rotation passes through the contact point ($v_{\text{contact}} = 0$).

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "The angular momentum of a moving body remains constant, if..." **[Ans: N/A]**
- **[2025 Shift] Q#3 (MCQ):** "Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1 = 4 \text{ kgm}^2, r_1 = 20 \text{ cm}$, $I_2 = 20 \text{ kgm}^2$ and $r_2 = 30 \text{ cm}$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?

- (A) 6 h
- (B) 18 h
- (C) 24 h
- (D) 30 h

Correct Answer: None

Solution Analysis: When the earth contracts suddenly, no external torque is acting on it. So its angular momentum remains conserved. Angular momentum is $L = I\omega$. For a spherical body, $I \propto R^2$. If the radius becomes half, $R' = (R/2)$. So the new moment of inertia becomes $I' = I((1/2))^2 = (I/4)$. Using conservation of angular momentum, $I\omega = I'\omega'$. $I\omega = (I/4)\omega'$. $\omega' = 4\omega$. So the earth will rotate 4 times faster. Time period is inversely proportional to angular velocity, so $T' = (T/4)$. Present duration of a day is $T = 24 \text{ h}$. Hence, $T' = 6 \text{ h}$.

Benchmark Problem 2: KCET 2026 (Q#2)

The angular momentum of a moving body remains constant, if

- (A) net external force is applied
- (B) net pressure is applied
- (C) net external torque is applied
- (D) net external torque is not applied

Correct Answer: None

Solution Analysis: The angular momentum of a moving body remains constant if net external torque is not applied. So the correct option is: Option D: net external torque is not applied. According to the principle of conservation of angular momentum, if the net external torque acting on a body or system is zero, then its angular momentum remains constant. Mathematically, $\tau_{\text{ext}} = (dL/dt)$. If no net external torque acts, $\tau_{\text{ext}} = 0$. So, $(dL/dt) = 0$. This means angular momentum L does not change with time. Hence, the angular momentum remains constant when net external torque is not applied.